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SUPERSEDING
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(See 6.6)

MILITARY SPECIFICATION

PIPE, FITTINGS, AND ADHESIVE KITS, GLASS-REINFORCED THERMOSETTING EPOXY RESIN FOR SHIPBOARD PIPING SYSTEMS

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers glass-reinforced thermosetting epoxy resin pipe, glass-reinforced thermosetting epoxy resin fittings, and thermosetting epoxy resin joint adhesive kits.

1.2 Classification. The classification shall be as specified in 1.2.1 through 1.2.3.

1.2.1 Pipe. Glass-reinforced thermosetting epoxy resin pipe shall be of one type, manufactured by the filament-winding process with a pressure and temperature rating of not less than 200 pounds per square inch (lb/in^2) at 150 degrees Fahrenheit (°F). Pipe shall be of the dimensions as specified (see 3.4.1).

1.2.2 Fittings. Unless otherwise specified, glass-reinforced thermosetting epoxy resin fittings shall be of the following types and configurations, with a pressure and temperature rating of not less than 200 lb/in^2 at 150°F.

1.2.2.1 Types. Fittings shall be of the following types, as specified (see 6.2.1):

Type I - Compression molded.
Type II - Filament wound.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2.2 Configurations. Fittings shall be of the following configurations (see figures 1-21), nominal sizes and dimensions (see tables VI-XXIII), as specified (see 6.2.1):

	Figure	Table
Adapters: (to be used only where unavoidable)		
tapered socket X male straight thread	1	VI
tapered spigot X male straight thread	2	VII
Cap, hexagonal, threaded	3	VIII
Coupling	4	IX
Cross	5	X
Elbows:		
90-degree, short radius	6	XI
90-degree, long radius	7	XI
45-degree, short radius	8	XII
45-degree, long radius	9	XII
Flanges:		
Plain, 150 lb/in^2, ANSI B16.5 drilling	10	XIII
Blind, 150 lb/in^2, ANSI B16.5 drilling	10	XIII
Plain, MIL-F-20042, class 150/250 pound drilling	11	XIV
Lateral, 45-degree	12	XV
Plug	13	XVI
Cap, tapered end, male socket	14	XVII
Reducers:		
Bushing type	15	XVIII
Tapered body, concentric	16	XIX
Saddle, instrumentation	17	XX
Tees:		
Straight, short radius	18	XXI
Straight, long radius	19	XXI
Standard tapered socket	20	XXII
Standard tapered spigot	21	XXIII

1.2.3 Adhesive kits. Thermosetting epoxy resin joint adhesive kits shall be one of the following sizes, as specified in 3.2.3 (see 6.2.1):

Small
Medium
Large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

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SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665	- Boxes: Paperboard, Metal Edged and Components.
PPP-B-676	- Boxes, Setup.
PPP-C-96	- Cans, Metal, 28 Gauge and Lighter.

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MIL-P-116	- Preservation, Methods of.
MIL-F-20042	- Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-C-46081	- Coating Compound, Thermal Insulating (Intumescent).

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-H28/7	- Screw-Thread Standards for Federal Services. Section 7 Pipe Threads, General Purpose.
FED-STD-141	- Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.
FED-STD-313	- Material Safety Data Sheets Preparation and the Submission of.
FED-STD-595	- Colors.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-129	- Marking for Shipment and Storage.

2.1.2 Other Government documents and drawings. The following other Government document and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR

Code of Federal Regulations, Title 29, Part 1910, Section 2206 - Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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DRAWING

NAVAL SHIP SYSTEMS COMMAND (NAVSHIPS)

1385782 - Hanger, Pipe for Submarines.

(Copies of specifications, standards, drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.5 - Pipe Flanges and Flanged Fittings. (DoD adopted)

Y14.6 - Screw-Thread Representation. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 618 - Standard Methods of Conditioning Plastics and Electrical Insulating Materials for Testing. (DoD adopted)

D 1763 - Standard Specification for Epoxy Resins.

D 2105 - Standard Test Method for Longitudinal Tensile Properties of Reinforced Thermosetting Resin Pipe and Tube.

D 2517 - Standard Specification for Reinforced Epoxy Resin Gas Pressure Pipe and Fittings. (DoD adopted)

D 2584 - Standard Test Method for Ignition Loss of Cured Reinforced Resins. (DoD adopted)

D 2697 - Standard Test Method for Volume Nonvolatile Matter in Clear or Pigmented Coatings. (DoD adopted)

D 2924 - Standard Test Method for External Pressure Resistance of Reinforced Thermosetting Resin Pipe.

D 2992 - Standard Method for Obtaining Hydrostatic Design Basis for Reinforced Thermosetting Resin Pipe and Fittings.

D 3567 - Standard Practice for Determining Dimensions of Reinforced Thermosetting Resin Pipe (RTRP) and Fittings.

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

E 119 - Standard Methods of Fire Tests of Building Construction and Materials. (DoD adopted)

E 162 - Standard Test Method for Surface Flammability of Materials Using a Radiant Heat Energy Source. (DoD adopted)

E 662 - Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 2200 Mill Road, Alexandria, VA 22314.)

NATIONAL SANITATION FOUNDATION (NSF)
Standard No. 14 - Thermoset Plastic Pipe, Fittings, Valves, Tanks, Appurtenances, Joining Materials and Thermoset Plastic Coatings for Use in Potable Water Supply Systems.

(Application for copies should be addressed to the National Sanitation Foundation, Box 1468, Ann Arbor, MI 48106.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 General. The general requirements for pipe, fitting and adhesive kits shall be as specified in 3.2.1 through 3.2.3.

3.2.1 Pipe. Glass-reinforced thermosetting epoxy resin pipe shall be round and straight and of uniform density, resin content, and surface finish. The pipe shall be constructed by the filament winding process, using a thermosetting epoxy resin to impregnate strands of continuous E-type glass filament which are wound around a straight mandrel under controlled tension. E-type glass shall be wound at an angle of 54 3/4 ± 2 degrees to the pipe axis. The bore of the pipe shall contain a smooth, uniform liner to protect the glass fiber reinforcement. The liner shall contain a minimum of 80 percent thermosetting epoxy resin by weight and shall be reinforced with a compatible A-type or C-type glass. Liner thickness after curing shall be not less than 15 mils when measured in accordance with 4.5.2, and the total glass content for the

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reinforced pipe wall (not including liner) shall be not less than 65 percent by weight, when tested in accordance with 4.5.3. The exterior surface of the pipe shall be coated with an intumescent fire-retardant epoxy paint in accordance with 3.3.1.4.

3.2.2 Fittings. Glass-reinforced thermosetting epoxy resin fittings shall be round and of uniform density, resin content, and surface finish. They shall be fabricated from thermosetting epoxy resin and glass reinforcement similar to and compatible with that used in the construction of the pipe wall. The total glass content of the fittings shall be not less than 40 percent by weight (including liner, if applicable) when tested in accordance with 4.5.3. Exterior surfaces of fittings shall be coated with an intumescent fire-retardant epoxy paint in accordance with 3.3.1.4.

3.2.2.1 Type I construction. Type I fittings shall be constructed by the compression-molding process. They shall be reinforced with chopped glass fibers that shall be not less than 1/2 or more than 1 inch in length. The bore shall have a smooth, uniform surface. Protrusions of more than 1/16 inch of material into the internal fitting flow area shall be removed and blended with adjacent smooth surfaces.

3.2.2.2 Type II construction. Type II fittings shall be constructed by the filament-winding process. They shall contain a reinforced epoxy-resin liner of identical material and similar construction to that used in the pipe liner. The liner thickness shall be not less than 15 mils after curing.

3.2.3 Adhesive kits. Adhesive kits shall include at minimum, a base epoxy resin (adhesive part "A"), hardener (adhesive part "B"), joint cleaner (methylene chloride), applicator brushes, mixing spatulas, paper towels and instruction sheet. Adhesive kits shall be furnished in the following combined adhesive size ranges:

Small 1-1 1/2 ounces Medium 5-7 ounces Large 9-10 ounces

3.3 Materials. The material requirements shall be as specified in 3.3.1 through 3.3.1.4.

3.3.1 Pipe and fittings. Pipe and fittings material shall meet the requirements as specified in 3.3.1.1 and 3.3.1.4.

3.3.1.1 Glass composition. Glass material shall be E-type, A-type, or C-type glass having the composition by weight percent specified in table I. Glass fibers shall be treated with a sizing material containing a coupling agent to provide bonding compatibility with the resin system and to control processing properties.

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TABLE I. Glass composition.

Constituent	E-type (percent)	C-type (percent)	A-type (percent)
B_2O_3	5 - 10	4 - 8	0
CaO	16 - 25	11 - 16	7 - 9
Al_2O_3	12 - 16	3 - 6	1 - 3
SiO_2	52 - 56	58 - 68	70 - 80
MgO	0 - 5	1 - 4	1 - 4
Na_2O and K_2O	0 - 2	6 - 10	7 - 15
TiO_2	0 - 1.5	0 - 0.9	0 - 0.9
Fe_2O_3	0 - 0.8	0.05 - 0.50	0 - 0.05
F_2	0 - 1.0	----	0 - 1.0

3.3.1.2 Epoxy resin. Epoxy resin material shall be in accordance with ASTM D 1763 for type I, grade 1, class A, bisphenol A, epichlorohydrin epoxy resin. Epoxy resin shall be characterized by its epoxide equivalent, viscosity, and color in accordance with 4.5.4. The resin shall be of uniform quality and shall be protected from contamination during transport, storage, and composite formulation.

3.3.1.3 Curing agents. The epoxy resin shall be cured with an aromatic-amine-type hardener.

3.3.1.3.1 Additives. The resin-hardener system may contain a limited quantity of accelerators, pigments, dyes, ultraviolet degradation inhibitors, fire retardants, and viscosity control agents. When combined with the composite structure, these additives shall not interfere with the ability of the pipe or fittings to meet the performance requirements of this specification. The total quantity of these additives shall be not more than 10 percent by weight of the resin-hardener system. The additives shall not contain asbestos in any form (see 4.5.4).

3.3.1.4 Exterior intumescent coating. The exterior surfaces of pipe and fittings shall be coated to a smooth finish with an intumescent fire-retardant epoxy compound in accordance with MIL-C-46081. This two-component intumescent coating shall be mixed, activated, and applied in accordance with MIL-C-46081 and the coating manufacturer's instructions. The pipe coating shall be applied to a minimum dry thickness of 8 mils with a maximum average thickness of 15 mils. Fitting coatings shall have an average dry thickness of 10 ± 2 mils. If necessary to assure the proper adhesion of the coating, exterior surfaces of pipe and fittings may be cleaned with a compatible solvent or may be lightly sanded. No primers, tie-coats, topcoats, or other compounds in

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addition to the coating shall be applied. The coating shall not be applied to any interior, wetted, bonding, threaded, or sealing surfaces of pipe or fittings. The adhesion of the coating shall be sufficient to pass the test specified in 4.5.6. The coating shall be furnished in color number 26270, haze grey, in accordance with FED-STD-595, and coated pipe and fittings shall approximate the standard color chip provided in FED-STD-595.

3.3.2 Joint adhesive. Joint adhesive shall be as specified in 3.3.2.1 through 3.3.2.2.

3.3.2.1 Resin-hardener system. Pipe and fittings shall be joined with a compatible epoxy adhesive consisting of an epoxy novolac resin cured with an aliphatic amine-type hardener. The resin-hardener system may also contain additives such as accelerators, inhibitors, and viscosity control agents, provided that the total quantity of these additives does not exceed 18 percent by weight of the resin-hardener system. Cured joint adhesive shall have a minimum ultimate shear strength as specified in 3.6.1.5. The base adhesive resin and hardener shall have a working pot life of at least 15 minutes at 75°F after both components are properly mixed. Maximum cure times for the mixed adhesive system after application to bonding surfaces shall not exceed 24 hours at 60°F or 30 minutes at 225°F.

3.3.2.2 Adhesive kits. The joint cleaner of the adhesive kits shall be methylene chloride supplied in metal dispensing containers constructed to be resealed after opening and to limit spillage if accidentally turned over after opening. Epoxy resin and hardener shall be supplied in premeasured stoichiometric quantities ready for mixing. Each component shall be supplied in a metal container convenient for mixing and brush application. Each adhesive kit shall contain an information sheet to include mixing, cleaning, and bonding instructions, along with pot life, cure times, and appropriate safety precautions. Adhesive kits shall be supplied in three size ranges (see 1.2.3) to meet the various requirements of each installation.

3.4 Dimensions. The dimension requirements shall be as specified in 3.4.1 through 3.4.1.5.

3.4.1 Pipe. The nominal size and diameter for pipe shall be as specified in 3.4.1.1.

3.4.1.1 Nominal size and diameter. Pipe shall be in the nominal sizes, outside diameters, and wall thicknesses shown in table II. Measurements of the od shall exclude the thickness of the exterior intumescent coating (see 3.3.1.4).

3.4.1.2 Wall thickness. The pipe wall thickness shall be not less than that shown in table II. Pipe wall thickness shall not exceed the minimum fitting wall thickness (as specified in table III) for the same nominal pipe size (nps). Measurements of the pipe wall thickness shall exclude the thickness of the exterior intumescent coating (see 3.3.1.4).

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TABLE II. Pipe dimensions.

Nominal pipe size (nps)	Minimum od (inch)	Od tolerance (inch)	Minimum wall thickness (inch)
1	1.315	+0.060	0.062
1-1/2	1.900	+ .060	.072
2	2.375	+ .060	.108
3	3.500	+ .060	.108
4	4.500	+ .060	.113
6	6.625	+ .066	.129
8	8.625	+ .151	.173
10	10.750	+ .188	.200
12	12.750	+ .223	.250

3.4.1.3 Roundness. Differences between the major and minor od of the pipe at any one cross section shall not exceed 1-1/2 percent of the average od specified in 3.4.1.1.

3.4.1.4 Length. Pipe shall be furnished in random lengths of 15 to 20 feet. Each length of pipe shall be furnished with plain square cut ends or standard tapered ends (see table XXIII) and shall not contain jointers (two pieces of pipe coupled to make a standard length).

3.4.1.5 Squareness of cut. The departure from squareness of the end of any pipe shall not exceed 1/32 inch per inch of pipe diameter.

3.4.2 Fittings. Fittings shall meet the requirements as specified in 3.4.2.1 through 3.4.2.8.

3.4.2.1 Nominal sizes and detailed dimensions. Fittings shall be supplied in the nominal sizes and detailed dimensions shown on figures 1 through 21, along with tables VI through XXIII.

3.4.2.2 Standard tapered socket. Fitting sockets shall be machined and tapered in accordance with the dimensions, angles, and tolerances shown on figure 20 and table XXII. Eccentricity of the tapered socket relative to the fitting centerline shall not exceed 1/32 inch for fittings from 1 through 2-inch size, 1/16 inch for fittings from 3 through 6-inch size, and 3/32 inch for fittings from 8 through 12-inch size.

3.4.2.3 Wall thickness. The minimum wall thickness for each fitting shall not be less than the appropriate value shown in table III. For tapered sockets, this requirement shall apply to every section within 1/4 inch of the open end. The minimum wall thickness of fittings, such as tapered body concentric reducers, which connect two different pipe sizes shall be that applicable to the larger pipe size. For instrumentation saddles, the minimum thickness of the tapered side outlet shall be that applicable to the nominal size of the connecting branch pipe, while the minimum thickness of the rest of the saddle shall be that applicable to the nominal size of the header pipe to which the saddle is mounted.

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TABLE III. Minimum wall thickness of fittings.[1]

Nps (inches)	Minimum wall thickness (type I and II fittings) (inch)
1	0.18
1-1/2	.18
2	.18
3	.22
4	.22
6	.32
8	.30
10	.36
12	.42

3.4.2.4 Tolerances. Unless otherwise specified (see 6.2.1), the tolerance of fitting dimensions shall be plus or minus 1/16 inch for fitting sizes up to and including 6 inches and plus or minus 1/8 inch for fitting sizes 8 inches and larger, except for dimensions specifying bolt circle or bolt hole diameters, which shall be maintained within plus or minus 1/32 inch. Unless otherwise specified (see 6.2.1), the tolerance of angular dimensions expressed in degrees shall be plus or minus 1 degree. Measurement of fitting dimensions shall be in accordance with the procedure specified in 4.5.2.

3.4.2.5 Machined threads. Unless otherwise specified (see tables I or II and 6.2.1), machining of threads shall be in accordance with ANSI Y14.6 and FED-STD-H28.

3.4.2.6 Machined surfaces. Machined fitting surfaces, except tapered sockets, shall have a roughness height rating of 250 microinches or less.

3.4.2.7 Flanged fittings. Face-to-face and center-to-face dimensions of flanged fittings shall be in accordance with ANSI B16.5 for 150 lb/in^2 service.

3.4.2.8 Flange finish. Flange faces, shall be provided with raised sealing rings or serrations, as a minimum, between the bore of the flange and the inside edge of the bolt holes. A minimum of two sealing rings or raised serrations shall be concentric and suitable for service with full face gasketing.

3.5 Identification. Identification marking requirements shall be as specified in 3.5.1 through 3.5.3.

3.5.1 Pipe. Unless otherwise specified (see 6.2.1), each pipe shall be marked (after the exterior intumescent coating is applied and cured) legibly and permanently in black paint or ink stenciling that is free of lead, zinc, or sulfur, with the following constantly recurring information at intervals not greater than 3 feet:

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"Fiberglass/epoxy; Nominal pipe size _____
MIL-P-24608
Manufacturer's name or trademark _____
200 lb/in² at 150°F, maximum."

As an alternate, pressure-sensitive, grease-proof, waterproof, nonfading labels may be used. Markings on labels shall be of contrasting color to the label background.

3.5.2 Fittings. Unless otherwise specified (see 6.2.1), each fitting shall be marked (after the exterior intumescent coating is applied and cured) legibly and permanently in black paint or ink stenciling that is free of lead, zinc, or sulfur, with the following information:

"Nominal size _____
Type _____
MIL-P-24608
Manufacturer's name or trademark _____
200 lb/in² at 150°F, maximum."

As an alternate, pressure-sensitive, grease-proof, waterproof, nonfading labels may be used. Markings on labels shall be of contrasting color to the label background.

3.5.2.1 Flanges. In addition to the identification markings of 3.5.2 (see 6.2.1), each plain and blind flange drilled to 150 lb/in² in accordance with ANSI B16.5 standards (see figure 10) shall be distinguished with the additional marking:

"CML Drilling"

and each plain flange drilled to MIL-F-20042, class 150/250 pound requirements (see figure 11) shall be distinguished with the additional marking:

"NAVY Drilling"

3.5.3 Adhesive kits. Unless otherwise specified (see 6.2.1), each adhesive kit shall be marked as specified in 5.4.1.

3.6 Properties. Pipe, fittings, and adhesive kits shall meet the properties requirements as specified in 3.6.1 through 3.6.1.8.

3.6.1 Physical properties. The requirements for physical properties shall be as specified in 3.6.1 through 3.6.2.3.

3.6.1.1 Long-term cyclic pressure strength. The extrapolated long term hydrostatic strength of sample pipe after 150 x 10^6 pressure cycles at 150°F or higher shall be not less than 5000 lb/in² (hoop stress), when tested in accordance with 4.5.8.

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3.6.1.2 Cyclic pressure strength. Pipe, fittings, and adhesive bonded joints shall be cycled a minimum of 10,000 cycles with no signs of leakage or weepage when tested in accordance with 4.5.9.

3.6.1.3 Hydrostatic collapse strength. The minimum buckling scaling constant (K) for pipe shall be 5.1×10^5 at 150°F when tested in accordance with 4.5.10.

3.6.1.4 Ultimate tensile strength. The minimum longitudinal tensile strength $\left(\frac{PD}{4t}\right)$ for pipe shall be 8,500 lb/in² (axial stress) when tested in accordance with 4.5.11.

3.6.1.5 Ultimate shear strength. The minimum ultimate shear strength of cured joint adhesive shall be 1500 lb/in² when tested in accordance with 4.5.12.

3.6.1.6 Hydrostatic strength. Pipe, fittings, and adhesive bonded joints shall show no signs of leakage or weepage when pressurized at 400 lb/in² gauge and 800 lb/in² gauge in accordance with 4.5.13.

3.6.1.7 Hydrostatic pressure. Pipe and fittings shall show no signs of leakage, porosity, weepage, cracks, flaws, or other defects when subjected to a hydrostatic test using potable water or air at 300 lb/in² (1-1/2 times rated pressure) in accordance with 4.5.14.

3.6.1.8 Impact. The pipe shall show no sign of leakage or weepage when subjected to 2.26 foot-pounds of energy in accordance with 4.5.15.

3.6.2 Combustion properties. Pipe, fittings, and adhesive kits shall meet the requirements for combustion properties as specified in 3.6.2.1 through 3.6.2.3.

3.6.2.1 Surface flammability. The maximum flame spread index of pipe covered with the exterior intumescent coating specified in 3.3.1.4 shall not exceed a value of 25 when determined in accordance with 4.5.6.

3.6.2.2 Smoke density. The maximum specific optical density under flaming conditions for pipe covered with the exterior intumescent coating specified in 3.3.1.4 shall not exceed a value of 250 when determined in accordance with 4.5.17.

3.6.2.3 Fire resistance. The pipe assembly shall remain intact for a minimum of 20 minutes when tested in accordance with 4.5.18.

3.7 Material certification. The requirements for material certification shall be as specified in 3.7.1 through 3.7.3.1.

3.7.1 Chemical resistance. When specified in the contract or purchase order, information on the recommended long-term chemical resistance capabilities of pipe, fittings, or adhesive kits as specified herein shall be prepared (see 6.2.2).

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3.7.2 Potable water service. When specified in the contract or purchase order, data supporting the acceptability of pipe, fittings, and adhesive kits for potable water service as specified herein shall be prepared (see 6.2.2). Pipe, fittings and adhesive kits shall meet the requirements of NSF Standard No. 14 (relating to the use of thermosetting-plastic pipe, fittings, and joining materials in potable-water supply systems) and shall be approved and listed by NSF for potable water service. The manufacturer of products conforming to this specification shall be responsible for initiating and maintaining contracts with NSF for annual retesting and reapproval for potable water listing. Any change in raw materials or production processes that may bring about chemical changes in the materials used in the manufacture of products conforming to this specification shall require resubmission to the NSF by the manufacturer for toxicological evaluation. Pertinent data supporting potable water acceptability and results of NSF investigations shall be submitted by the manufacturer to the Naval Medical Command (NAVMEDCOM) for approval as specified herein.

3.7.3 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.7.3.1 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is form OSHA-20, found in and part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.4).

3.8 Workmanship. Pipe and fittings shall be free from indentations, delaminations, bubbles, pinholes, foreign inclusions, cracks, porosity, resin rich areas, and resin starved areas which, because of their nature, degree, or extent, detrimentally affect the strength and serviceability of the pipe and fittings. Pipe and fittings shall be uniform in color, surface finish, density, and other physical properties. Removal of surface imperfections by methods such as grinding or filing shall be permissible, providing the grinding or filing does not cut into any of the reinforcement in the pipe or remove more than 10 percent of the wall thickness of any pipe or fitting, and the required minimum wall thickness is maintained. Ground or filed areas in pipe and fittings shall be coated before shipment with the same base resin and intumescent fire-retardant epoxy compound as was used in the construction of the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification in accordance with the data ordering documents (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in table IV. When specified in the contract or purchase order, test reports shall be prepared (see 6.2.2).

TABLE IV. First article tests.

Examination or test	Requirement	Test	Number of samples
Pipe			
Liner thickness	3.2.1	4.5.2	Three each: 3, 6 and 12 nps
Total glass content	3.2.1	4.5.3	
Thickness, intumescent coating	3.3.1.4	4.5.5	
Adhesion, intumescent coating	3.3.1.4	4.5.6	
Dimensions	3.4.2	4.5.2	
Visual	3.5-3.5.3, 3.7.1-3.7.3.1	4.5.7	
Long-term cyclic pressure strength	3.6.1.1	4.5.8	18 (minimum); including 3, 6 and 12 nps
Cyclic pressure strength	3.6.1.2	4.5.9	See 4.5.9
Hydrostatic collapse strength	3.6.1.3	4.5.10	Five each: 3, 6 and 12 nps
Ultimate tensile strength	3.6.1.4	4.5.11	
Hydrostatic strength	3.6.1.6	4.5.13	See 4.5.13
Impact	3.6.1.8	4.5.15	One each: 3, 6 and 12 nps
Surface flammability	3.6.2.1	4.5.16	Four, prepared from sections of 3 nps
Smoke density	3.6.2.2	4.5.17	

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TABLE IV. First article tests. - Continued

Examination or test	Requirement	Test	Number of samples
Pipe - Continued			
Fire resistance	3.6.2.3	4.5.18	One each: 1, 2, 4 and 8 nps
Chemical resistance	3.7.1		Data requirements
Potable water service	3.7.2		Data requirements
Toxicity	3.7.3		Data requirements
Fittings			
Total glass content	3.2.2	4.5.3	1 each: 3, 6 and 12 nps
Thickness, intumescent coating	3.3.1.4	4.5.5	
Adhesion, intumescent coating	3.3.1.4	4.5.6	
Dimensions	3.4.2	4.5.2	3 each: 3, 6 and 12 nps
Visual	3.5-3.5.3, 3.7.1-3.7.3.1	4.5.7	
Cyclic pressure strength	3.6.1.2	4.5.9	See 4.5.9
Hydrostatic strength	3.6.1.6	4.5.13	See 4.5.13
Fire resistance	3.6.2.3	4.5.18	See pipe requirements
Chemical resistance	3.7.1		Data requirements
Potable water service	3.7.2		Data requirements
Adhesive kits			
Ultimate shear strength	3.6.1.5	4.5.12	See 4.5.12
Fire resistance	3.6.2.3	4.5.18	See pipe requirements
Chemical resistance	3.7.1		Data requirements
Potable water service	3.7.2		Data requirements
Toxicity	3.7.3		Data requirements

4.3.1 Samples for first article inspection. Pipe, fittings, and adhesive kits of each type, configuration, and size, as applicable, for which first article inspection is required shall be submitted in accordance with table IV.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in table V. Quality conformance inspection shall be performed on every lot of pipe, fittings, and adhesive kits acquired under this specification. When specified in the contract or purchase order, test reports shall be prepared (see 6.2.2).

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TABLE V. Quality conformance tests.

Examination or test	Requirement	Test	Inspection level and AQL
Pipe			
Thickness, intumescent coating	3.3.1.4	4.5.5	Level S-1, 6.5 percent
Dimensions	3.4.1.1-3.4.1.5	4.5.2	Level S-3, 4 percent
Visual	3.5-3.5.3 3.7.1-3.7.3.1	4.5.7	100 percent inspection[1]
Hydrostatic pressure test	3.6.1.7	4.5.14	Level S-3, 4 percent
Fittings			
Dimensions	3.4.2	4.5.2	Level S-3, 4 percent
Visual	3.5-3.5.3 3.7.1-3.7.3.1	4.5.7	100 percent inspection[1]
Hydrostatic pressure test	3.6.1.7	4.5.14	Level S-3, 4 percent
Adhesive kits			
Contents	3.2.3	4.5.7	Level S-3, 4 percent
Ultimate shear strength	3.6.1.5	4.5.12	Level S-1, 2.5 percent
Raw materials			
Glass	3.3.1.1	4.5.4	Level S-3, 2.5 percent
Epoxy resin	3.3.2.1	4.5.4	
Intumescent coating compound	3.3.1.4	4.5.4	

4.4.1 Lots for quality conformance inspection. The definitions of inspection terms used herein shall be in accordance with MIL-STD-109. For purposes of this specification, the following shall also apply:

4.4.1.1 Lot of pipe. Pipe of the same size constructed at essentially the same time and submitted for inspection at one time shall be considered a lot.

4.4.1.2 Lot of fittings. Fittings of the same type, kind, and nominal size constructed at essentially the same time and submitted for inspection at one time shall be considered a lot.

4.4.1.3 Lot of adhesive kits. Adhesive kits of the same size constructed at essentially the same time from the same lot of raw material and submitted for inspection at one time shall be considered a lot.

4.4.1.4 Lot of raw material. Each shipment of epoxy resin, glass, or intumescent coating compound received from a vendor by a pipe, fittings or adhesive contractor at the same time shall be considered a lot.

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4.4.2 Sampling for quality conformance inspection. Unless otherwise specified (see 6.2.1), a random sample of pipe, fittings, adhesive kits, and raw materials shall be selected from each lot in accordance with MIL-STD-105, using the inspection level and acceptable quality level (AQL) as specified in table V.

4.4.3 Nonconforming inspection lots. The disposition of inspection lots found unacceptable shall be in accordance with MIL-STD-105.

4.5 Examinations and test methods. Pipe, fittings, and adhesive kits shall meet the examination and test methods specified in 4.5.1 through 4.5.19.

4.5.1 Inspection conditions. Unless otherwise specified in this specification, all examinations and tests shall be conducted at ambient temperature and humidity. Unless otherwise specified (see 6.2.1), specimens that require conditioning shall be conditioned in the Standard Laboratory Atmosphere in accordance with procedure A of ASTM D 618 for not less than 48 hours before testing. Examinations and tests, except for those specified in 4.5.5, 4.5.6, 4.5.16 and 4.5.17, shall be conducted before the application of the exterior intumescent coating to facilitate visual detection of defects, leakage, and structural damage.

4.5.2 Dimensional examination. Dimensions and tolerances of pipe, fittings, and tapers shall be determined in accordance with ASTM D 3567.

4.5.3 Total glass content. Total glass content of sample pipe and fittings shall be determined in accordance with ASTM D 2584. For purposes of this specication, total glass content by weight percent shall be defined as 100 minus the ignition loss (expressed in weight percent).

4.5.4 Raw materials. Each lot of epoxy resin, curing agents, joint cleaner, glass, and intumescent coating compounds shall be certified to be in accordance with the material specifications of 3.3. Epoxy resin used in pipe and fitting construction shall be characterized by its epoxide equivalent, viscosity, and color which shall be determined in accordance with the standard test methods specified in ASTM D 1763. When specified in the contract or purchase order, records of material certification shall be maintained and supplied upon request to the contracting activity (see 6.2.2).

4.5.5 Intumescent coating thickness. Average thickness of intumescent coating compound applied to sample pipe and fittings shall be determined using procedures similar to those specified in ASTM D 2697. The coating shall be allowed to cure on the sample pipe or fittings for at least 7 days in the Standard Laboratory Atmosphere before thickness measurements are made. Thickness shall be measured from the apparent interface between the pipe and coating outward to the edge of the coating. Four measurements, 90 degrees apart, shall be made at three separate cross sections of the pipe or fitting, separated by no less than five nominal diameters of length for pipe cross sections, and one nominal diameter of length for fitting cross sections. The average value of the twelve coating thickness measurements shall be used to determine conformance to the thickness requirement specified in 3.3.1.4.

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4.5.6 Adhesion of intumescent coating. Adhesion of the intumescent coating to the sample pipe and fittings shall be determined in accordance with method 6304 (knife test) of FED-STD-141. The coating shall be allowed to cure on the sample pipe or fitting for at least 7 days in the Standard Laboratory Atmosphere before adhesion tests are made. After curing, the intumescent coating shall be difficult to furrow off the pipe or fitting with the knife and shall not flake, chip, or powder. The knife cut shall show beveled edges.

4.5.7 Visual examination. Pipe and fittings meeting this specification shall be visually examined to verify conformance with 3.5 and 3.7. Any pipe or fitting having one or more of the defects listed shall be considered a defective unit. Samples of adhesive kits selected in accordance with tables IV and V shall be examined to verify conformance with the requirements of 3.3.2.2 and 3.5.3. Missing or incorrect components, instructions, precautions, quantities by weight or number, or identification markings shall be considered defects and any sample adhesive kit having one or more of the defects listed shall be considered a defective unit.

4.5.8 Long term cyclic pressure strength. A minimum of eighteen specimens of pipe shall be exposed to cyclic internal pressures at 150°F or higher in accordance with procedure A of ASTM D 2992.

4.5.9 Cyclic pressure strength. Test specimens shall include a tee, a flange and one other fitting selected by the contractor. The fitting shall be joined to pipe sections at least 18 inches long or three diameters, whichever is longer, using the joining method, construction and adhesion intended for field assembly. Specimen diameters for testing in each configuration shall include the maximum product size for each method of construction. In addition, the following other sizes shall be tested as follows:

Product size range	Test specimen size
1, 1-1/2, 2, 3	3
4, 6	6
8, 10, 12	12

In each range of size the contractor may select to test a smaller product size. Each assembly shall be cycled between 0 and 400 lb/in^2 gauge for a minimum of 10,000 cycles without failure. Failure shall be classified as any leakage or weeping from the pipe, fitting, or joint.

4.5.10 Hydrostatic collapse strength. The external pressure resistance of sample pipe shall be determined at 150°F in accordance with ASTM D 2924. The buckling scaling constant "K" shall be calculated as defined in ASTM D 2924.

4.5.11 Ultimate tensile strength. The minimum longitudinal tensile strength of sample pipe shall be determined in accordance with ASTM D 2105.

4.5.12 Ultimate shear strength. Ultimate shear strength of the adhesive used to bond pipe and fittings shall be determined in accordance with the procedure and calculations described in ASTM D 2517.

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4.5.13 Hydrostatic strength. Test specimens shall include a tee, a flange and one other fitting selected by the contractor. These shall be joined to pipe sections at least 18 inches long or three diameters, whichever is longer, using the joining method, construction, and adhesion intended for field assembly. Specimen diameters for testing in each configuration shall include the maximum product size for each method of construction. In addition, the following other sizes shall be tested as follows:

Product size range	Test specimen size
1, 1-1/2, 2, 3,	3
4, 6	6
8, 10, 12	12

In each range of size the contractor may select to test a smaller product size to qualify only the smaller size. Each assembly shall be subjected to a pressure of 400 lb/in^2 for 5 minutes, followed by a pressure drop to 0 lb/in^2 gauge and a repressurization to 800 lb/in^2 gauge for a minimum of 5 minutes.

4.5.14 Hydrostatic pressure test. Each sample pipe or fitting shall be subjected at room temperature to an internal pressure of 300 lb/in^2 as specified in 3.6.1.7. The test pipe or fitting may be tested with air or water. If proof tested with air, the sample shall be immersed in water and pressure shall be held for at least 1 minute while the specimen is observed for leakage. If proof tested with water, all air shall be vented from the test specimen, which shall be completely filled with fresh water and pressurized to 300 lb/in^2 for at least 5 minutes and then examined for porosity while still under pressure.

4.5.15 Impact. A section of pipe shall be mounted on pipe hangers spaced at a distance of 3 feet. One-quarter inch neoprene padding may be used as hanger padding. The pipe shall be empty and under no pressure. A 1-1/4 inch steel ball, weighing 0.282 pounds, shall be dropped from a distance of 8 feet in order to impact squarely at the top of the pipe and at center span. This impact imparts approximately 2.26 foot-pounds of energy to the pipe. Other ball sizes of different weights up to a 3-inch ball diameter may be used, and the drop height adjusted accordingly to achieve an equivalent impact energy. After impact, the pipe shall be filled with water and pressurized to 200 lb/in^2 gauge for a minimum of 2 minutes. During this time, the pipe shall be inspected for evidence of weepage. Evidence of water on the pipe exterior shall be considered a failure. The above test shall be repeated for a total of four impacts at center span by rotating the pipe 90 degrees before each new impact.

4.5.16 Surface flammability. The maximum flame spread index of sample pipe coated with intumescent paint shall be determined in accordance with ASTM E 162. Specimens for these tests shall consist of three 110-1/2 ± 1/2 degree sections (approximately 2 inches wide) of 2-inch nps, mounted in the standard 6 by 18-inch steel frame so that the coated exterior surfaces of the pipe sections face the radiant panel.

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4.5.17 Smoke density. The maximum specific optical density under flaming conditions for sample pipe coated with intumescent paint shall be determined in accordance with ASTM E 662. Specimens for these tests shall consist of three 48-1/2 ± 1/2 degree sections (approximately 1 inch wide) of 2-inch nps, mounted in the standard 3-inch square specimen holder so that the coated exterior surfaces of the pipe sections face the radiant heat furnace.

4.5.18 Fire resistance. A spool section shall be prepared which consists of the contractor's standard pipe and a single fitting located at the center of the spool length. The fitting may be a coupling, 90 degree elbow, 45 degree elbow, tee, cross or 45 degree lateral. The fitting shall be the thinnest wall thickness that is available from the contractor, but which meets the minimum wall thickness criteria of 3.4.2.3. The spool piece shall be flanged at both ends, with a distance of 3 feet plus or minus 6 inches from the fitting joint to the flange joint. If a fitting with more than two joints, such as a tee or cross is used, additional pipe legs and flanges may be added in accordance with the above joint-to-joint dimensions. Blind flanges shall be used where necessary. The pipe assembly shall be supported with the fitting centered horizontally above a fire tray with minimum dimensions of 3 by 3 feet. Steel split ring hangers shall be used to support the pipe, similar to type 1 or 2 hangers as shown in Drawing 1385782. Hangers may be suspended or supported from the ground. The hangers shall allow the assembly to move freely in the horizontal plane and permit axial loading of each of the bonded joints. This may be done by loosening one or more hanger clamps, or in the case of a ground supported hanger, eliminating the top strap entirely. Hangers shall be mounted with the following spans:

Nps (inches)	Hanger span (feet)
1	4.5
2	5.0
4	7.0
8	9.0

The above distances are from hanger center to hanger center via the centerline of the pipe assembly, including fitting, between the two hangers. For crosses, the above distances shall apply to the hangers that are in direct line with one another. For branched assemblies, such as tees or laterals, the hanger supporting the branch shall be within 3 inches of the flange at the end of the support branch. Pipe joints shall be constructed with adhesive kits and tapers in accordance with this specification. The test spool shall be filled with fresh water pressurized to 200 ± 5 lb/in^2, and exposed to a hydrocarbon fuel fire with the fitting centered above the fire. The test shall be run under stagnant water conditions and pressure maintained on the assembly for the duration of the test. The pipe, bonded joints, and fittings shall not be coated or otherwise insulated from the fire in any way. However, the flanges shall be sufficiently insulated to prevent leakage at the flange gasket or bolt holes. Flame temperature shall be in accordance with ASTM E 119 and shall be continuously measured by means of two thermocouples in the horizontal plane on either side of the fitting. The average of the two temperature measurements shall be taken as the flame temperature. The average flame temperature curve, expressed in degree-minutes, should be equal to or greater than that specified in ASTM E 119. The fire test shall be run for 20 minutes or until pipe, fitting, or joint failure,

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whichever comes first. Following the 20-minute fire test, the test spool shall remain pressurized to 200 ± 5 lb/in^2 gauge for a minimum of 10 minutes. During this time the total leakage from the pipe, fitting, and bonded joints shall be measured. Failure shall be defined as an average leakage rate greater than 2 percent of the maximum rated flow of the pipe, based on a maximum flow velocity of 17 feet per second.

4.5.19 Toxicity. The contractor shall have the toxicological formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.6 Inspection of packaging. Sampling and inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with this specification. Preservation-packaging requirements not identified to specifications shall be examined for the defects listed below on the basis of the sampling procedures and inspection levels specified in MIL-P-116. Packing and marking requirements of section 5 not identified to specifications shall be examined for the defects listed in 4.6.1.1 in accordance with MIL-STD-105, using an AQL of 4.0 percent defective.

4.6.1 Classification of defects. Classification of defects shall be as specified in 4.6.1.1 through 4.6.1.3.

4.6.1.1 Preservation-packaging. Preservation-packaging examination, defects shall be classified as follows:

	Defect	Paragraph
Critical:	None	
Major:		
101	End protection not provided, improper material and method	5.2.1
102	Bundle ends not protected, improper materials and method; missing straps, platform not provided	5.2.1.1(a)
103	Cans not as specified, inner seal missing, not a leakproof closure, coatings not applied, side seam not striped	5.2.1.3(a) and (b)
104	Inner packing forms missing	5.2.1.3.1
Minor:		
201	Lengths per package not as specified	5.2.1.1
202	Bag not closed	5.2.1.3(c)

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	Defect	Paragraph
Critical:	None	
203	Plastic brush cover missing	5.2.1.3(d)
204	Instruction sheet missing	5.2.1.3(e)

4.6.1.2 Packing. Packing examination, defects shall be classified as follows:

	Defect	Paragraph
Critical:	None	
Major:		
101	Nonconforming container, closures, sealing, and reinforcing	5.3.1 and 5.3.2
Minor:	None	

4.6.1.3 Marking. Marking examination, defects shall be classified as follows:

	Defect	Paragraph
Critical:	None	
Major:		
101	Marking missing, not clear	5.4.1
102	Caution or identification markings missing, not clear	5.4.1
Minor:	None	

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Pipe and fitting open-end protection. Open ends of each pipe and fitting shall be protected by nontransparent plastic caps, plugs, or wraps (at the contractor's option) to prevent damage and degradation of critical and machined surfaces. Plastic caps, plugs or wrap material shall incorporate ultraviolet light protection by use of carbon black or other equally efficient pigments or inhibitors and shall be compatible with the item protected. Wrap materials shall be a minimum of 6 mils thick and secured to the item with waterproof, grease-proof, pressure-sensitive tape.

5.2 Preservation-packaging. Preservation-packaging shall be level A, C, or commercial as specified (see 6.2.1).

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5.2.1 Level A.

5.2.1.1 Pipe. Pipe shall be packaged as specified below. Bundles shall contain only one pipe size and length (see table II and 3.4.1.4).

Size (inches)	Package	Maximum lengths per package
1	Bundle	15
1-1/2	Bundle	10
2	Bundle	8
3	Bundle	6
4	Bundle	6
6	Bundle	2
8	Bundle	2
10	Each or unit	1
12	Each or unit	1

(a) Bundles. Bundle ends shall be wrapped or capped with a waterproof treated fiberboard. The wrap or cap shall completely enclose each end, extend approximately 6 inches onto the bundle, and be secured with 1/2-inch wide, corrosion-resistant treated strapping. Each bundle (except for 7-1/2 foot lengths) shall be provided with a centered strap, in addition to the two end straps specified above. To protect the pipe coating from damage, a complete circumferential fiberboard or equal material wrap shall be placed between the pipe and strap. To enable handling, each bundle shall be secured to a wood or plywood constructed platform that shall also incorporate provisions for material handling equipment and devices.

(b) Unit pipe package. At the contractor's option, each pipe shall be placed in a container constructed of fiberboard, wood, or plywood. Fiberboard containers shall be in accordance with PPP-B-636 or PPP-B-640; wood and plywood containers shall be in accordance with PPP-B-621 and PPP-B-601, respectively. Except as specified in 5.3, selection of container type, variety, grade, and style shall be at the contractor's option. The interior (sides, ends, top, and bottom) of each wood box shall be lined with fiberboard pads or inserts to provide a snug-fitting, non-shifting load to prevent damage to the pipe coating.

5.2.1.2 Fittings. Unless otherwise specified (see 6.2.1), fittings shall be individually packaged, except when used in pairs or sets. When packaged in pairs or sets, each fitting shall be enclosed in a wrap or bag or shall be cushioned to prevent coating damage from contact with adjacent parts. Fittings shall be placed in a folding or set-up paperboard, fiberboard, wood, or plywood box. Paperboard boxes shall be in accordance with PPP-B-566, PPP-B-665, or PPP-B-676; fiberboard boxes shall be in accordance with PPP-B-636, wood and plywood boxes shall be in accordance with PPP-B-621 and PPP-B-601. Selection, except as specified in 5.3, of the type, variety, grade, style, class, and closure shall be at the contractor's option. The gross weight of paperboard boxes shall not exceed 10 pounds. The gross weight of other boxes shall not exceed the weight limitations of the box specification.

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5.2.1.3 Adhesive kits. Adhesive kit components (see 3.3.2.2) shall be unit protected as follows:

(a) Cleaner and hardener. The can for each component shall be in accordance with PPP-C-96, type VIII, round, and shall be fitted with a metal screw cap closure. Each cap shall be provided with a liner, pad, and facing. The facing shall be made of material that will provide a leakproof seal. The can containing the cleaner shall be provided with a metal snug-fitting, press-fit inner seal and an additional spill-proof, dispensing type, double-cap spout. Can and screw cap materials shall be at the contractor's option. Protective coatings shall be applied to the interior and exterior can surfaces. In addition, the side seam shall be striped with a corrosion-resistant coating.

(b) Adhesive. The can shall be in accordance with PPP-C-96, type V, round, class 2. Other can requirements, except for screw cap closure, liner, pad, facing, inner seal, and spout, shall be as specified in 5.2.1.3(a).

(c) Towels. Towels shall be folded or rolled and enclosed in a plastic bag. Bag closure shall be accomplished by heat sealing or by providing a press-fit or equal type closure.

(d) Brushes. The bristle end of each brush shall be enclosed in a plastic cover to protect the bristle and maintain brush contour.

(e) Instruction sheet. The instruction sheet shall be included in the towel package.

(f) Spatulas. Spatulas shall require no packaging.

5.2.1.3.1 Components of each kit shall be contained in a fiberboard box in accordance with PPP-B-636, type CF, class weather-resistant. The variety, grade, and style shall be at the contractor's option. Box closure, sealing, and reinforcing shall be in accordance with method V of the appendix to the box specification. Box loading shall be in accordance with the appendix to the box specification. The box shall be fitted with interior fiberboard packing forms, such as partitions or dividers, to provide separate cells for each can.

5.2.2 Level C. Piping and fittings shall be protected as specified in 5.2.1. Adhesive kit components and kits shall be packaged in a manner to provide protection against damage during shipment from the supply source to the first receiving activity for immediate use. When such methods meet the requirements of this level, the contractor's normal packaging methods may be used.

5.2.3 Commercial. Piping, fittings, and adhesive kits shall be preserved as specified in ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2.1).

5.3.1 Level A. Pipe, fittings, and adhesive kits shall be packed level A as specified in 5.3.1.1 through 5.3.1.3.

5.3.1.1 Pipe. Level A packing for pipe shall be as specified in 5.2.1.

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5.3.1.1.1 Bundles. Before securing to the wood or plywood platform, each bundle shall be completely wrapped and sealed with a waterproof, flexible material. Further packing shall not be required.

5.3.1.1.2 Unit pipe packages. Unit pipe packages, as specified in 5.2.1.1(b), shall be as follows:

(a) Fiberboard shall be of the weather-resistant class. Boxes shall be closed, sealed, and reinforced in accordance with the appendix to the applicable box specification, except that method V shall apply for boxes in accordance with PPP-B-636.
(b) Wood and plywood boxes shall be in accordance with class 2 or the overseas type as applicable. Box closure and reinforcing shall be in accordance with the appendix to the applicable box specification.

5.3.1.2 Fittings. Paperboard and fiberboard packages shall be packed for shipment in containers in accordance with any one of the following specifications at the contractor's option. Wood or plywood containers shall need no further packing, but shall be in accordance with the requirements specified herein.

Specification	Container	Type or class
PPP-B-601	Wood, cleated-plywood	Overseas type
PPP-B-621	Wood, nailed and lock corner	Overseas, class 2
PPP-B-585	Wood, wirebound	Class III

Box closure shall be in accordance with the appendix to the applicable box specification. The gross weight of a shipping container shall be in accordance with the applicable box specification. Shipping containers exceeding 200 pounds shall be modified by the addition of skids in accordance with the applicable box specification.

5.3.1.3 Adhesive kits. Adhesive kits shall be packed for shipment in containers as specified in 5.3.1.2 or containers in accordance with PPP-B-636, type CF, class weather-resistant, or PPP-B-640, class 2. The variety, grade, and style shall be at the contractor's option. Box closure, sealing, and reinforcing shall be in accordance with the appendix to the applicable box specification, except that method V shall apply for boxes in accordance with PPP-B-636.

5.3.2 Level B. Pipe, fittings, and adhesive kits shall be packed level B as specified in 5.3.2.1 through 5.4.1.1.2.

5.3.2.1 Pipe. Level B packing for pipe shall be as specified in 5.3.2.1.1 and 5.3.2.1.2.

5.3.2.1.1 Bundles. Packing shall be as specified in 5.3.1.1.1.

5.3.2.1.2 Unit pipe packages. Packing shall be as specified in 5.3.1.1.2, except as follows:

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(a) Fiberboard boxes shall be of the domestic grade or class. Boxes shall be closed in accordance with the appendix to the applicable box specification, except that method I shall apply for boxes in accordance with PPP-B-636.
(b) Wood and plywood boxes shall be in accordance with class 1 or domestic type.

5.3.2.2 Fittings. Packing shall be as specified in 5.3.1.2, except that wood and plywood boxes shall be of the domestic type and wirebound, class II.

5.3.2.3 Adhesive kits. Packing shall be as specified in 5.3.1.3, except that boxes shall be of the domestic type or class, as applicable. Fiberboard boxes shall be closed as specified in 5.3.2.1.2(a).

5.3.3 Level C. Piping, fittings, and adhesive kits shall be packed in a manner acceptable to the common carrier which shall ensure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Container, packaging, and method of shipment shall be in accordance with Uniform Freight Classification Ratings, Rules and Regulations and National Motor Freight Classification or other carrier rules as applicable to the mode of transportation.

5.3.4 Commercial. Pipe, fittings, and adhesive kits shall be packed in accordance with ASTM D 3951.

5.4 Marking. In addition to any special marking required (see 6.2.1 and 5.4.1), unit packages, bundles, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 or ASTM D 3951. When specified (see 6.2.1), bar code marking of MIL-STD-129 shall apply.

5.4.1 Special marking. In addition to the markings of 5.4, the following shall apply:

5.4.1.1 Adhesive kits. The marking for adhesive kits shall be as specified in 5.4.1.1.1 through 5.4.1.1.2.

5.4.1.1.1 Cans. Each can shall contain the contractor's name and address and the following precautionary information:

"WARNING
Use with adequate ventilation. Avoid breathing vapor. Avoid contact with skin. Wash thoroughly after handling and before eating or smoking. No smoking in area of use. Do not use in the general vicinity of hot surfaces, open flame, or arc welding. Heat or ultra violet radiation may cause the formation of toxic by-products."

Each can of adhesive (epoxy resin) shall be identified as "Adhesive (epoxy resin), Part A." Each can of hardener shall be identified as "Hardener (adhesive), Part B." Each can of joint cleaner shall be identified as "Cleaner (methylene chloride)."

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5.4.1.1.2 Fiberboard box. Each box containing an adhesive kit shall be marked with the name and kit size, MIL-P-24608, date of manufacture, storage requirement, an itemized list of box contents, and the contractor's name and address.

6. NOTES

6.1 Intended use. Pipe, fittings, and adhesive kits covered by this specification are intended for service up to 150°F and 200 lb/in² in shipboard piping systems of 1 through 12-inch diameters conveying nonflammable and noncombustible fluids. Higher temperature service in systems constructed up to 200°F is permitted, provided constructed pressure does not exceed 100 lb/in².

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of fittings required (see 1.2.2.1).
(c) Configuration of fittings desired (see 1.2.2.2).
(d) Size range of adhesive kits required (see 1.2.3).
(e) First article requirements (see 3.1).
(f) If tolerances of fitting dimensions or angular dimensions are other than as specified (see 3.4.2.4).
(g) If fitting threads are other than specified (see 3.4.2.5).
(h) Additional markings, if required (see 3.5.1, 3.5.2, 3.5.2.1 and 3.5.3).
(i) If quality conformance sample is other than as specified (see 4.4.2).
(j) If test specimen conditioning is other than as specified (see 4.5.1).
(k) Level of packaging and packing required (see 5.2 and 5.3).
(l) If fittings are not to be individually packaged (see 5.2.1.2).
(m) Special marking requirements (see 5.4).
(n) If bar code marking is required (see 5.4).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.7.1, 3.7.2 and 4.5.4	Certificate of compliance	DI-E-2121	----
4.1.2	Inspection system program plan	DI-R-4803	----

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Paragraph no.	Data requirement title	Applicable DID no.	Option
4.3	First article inspection report	DI-T-4902	----
4.4	Reports, test	DI-T-2072	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of the number of units specified. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets (MSDS) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.5 Subject term (key word) listing.

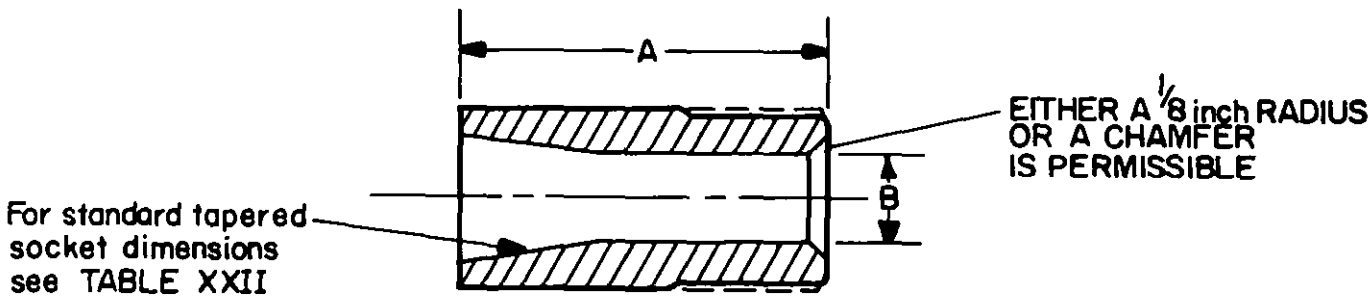
Adhesive kits
Filament-winding process
Glass filament
Glass-reinforced thermosetting epoxy resin
Intumescent coating
Methylene chloride
Stoichiometric

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4710-N845)

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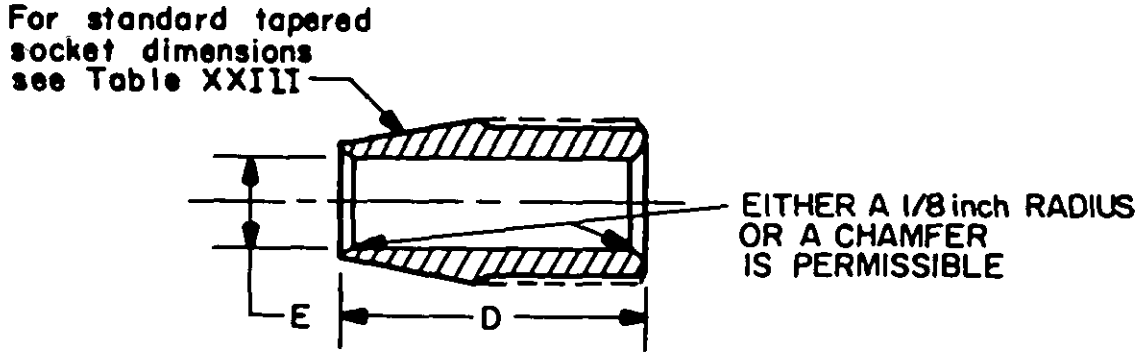
FIGURE 1. Standard adapter, tapered socket by male straight thread, type I or II.[1]

TABLE VI. Dimensions of tapered socket by male straight adapter. Threaded ends shall be in accordance with FED-STD-H28.[2]

	A	B
Nps	Overall length, minimum (inches)	Minimum internal diameter (inches)
1	3.250	0.875
1-1/2	3.250	1.500
2	4.000	1.890

[1] For tolerances, see 3.4.2.4.
[2] Adapters are also available in pipe thread configuration in accordance with FED-STD-H28/7.

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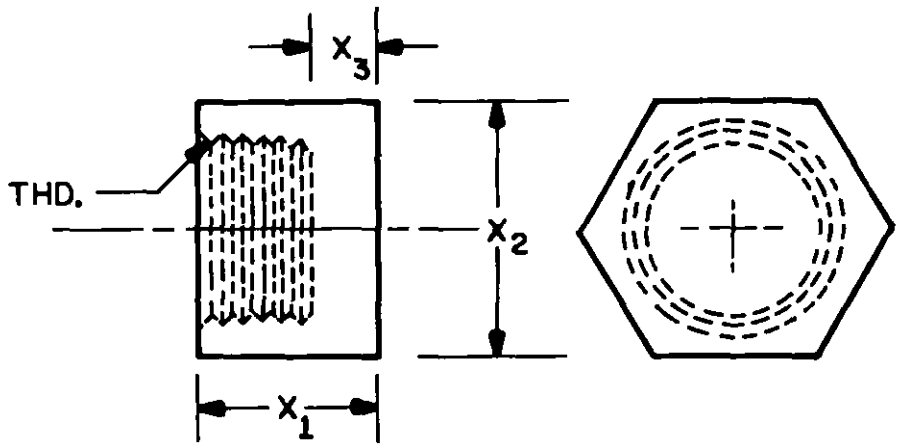
FIGURE 2. Standard adapter, tapered spigot by male straight thread, type II.[1]

TABLE VII. Dimensions of tapered spigot by male straight thread adapter. Threaded ends shall be in accordance with FED-STD-H28.[2]

	D	E
Nps	Overall length, minimum (inches)	Minimum internal diameter (inches)
1	2.375	0.875
1-1/2	2.500	1.500
2	3.500	2.000

[1] For unspecified tolerances, see 3.4.2.4.
[2] Adapters are also available in pipe thread configuration in accordance with FED-STD-H28/7 for systems up to 50 lb/in^2.

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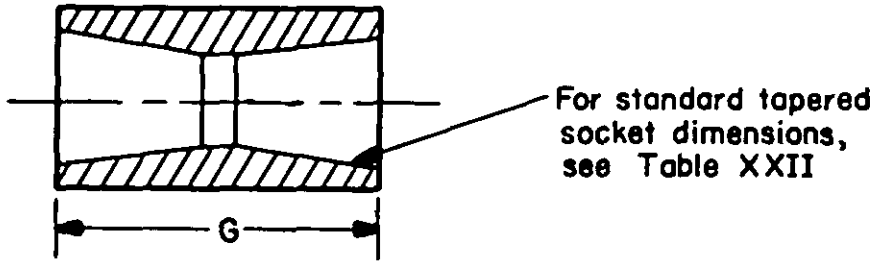
FIGURE 3. Hexagonal cap, threaded, type I.[1]

TABLE VIII. Dimensions of hexagonal threaded cap. Threads shall be in accordance with FED-STD-H28.[2] [3]

Nps	Overall length X_1 (inches)	Width across hex flats X_2 (inches)	Width across hex flats, tolerance (inch)	Minimum thickness X_3 (inch)
1	1.500	1.688	+0 -1/32	0.438
1-1/2	1.750	2.438	+0 -1/16	.500
2	2.000	3.000	+0 -1/16	.563

[1] May be used with type I or II fittings.
[2] For unspecified tolerances, see 3.4.2.4.
[3] Caps are also available in pipe thread configurations in accordance with FED-STD-H28/7.

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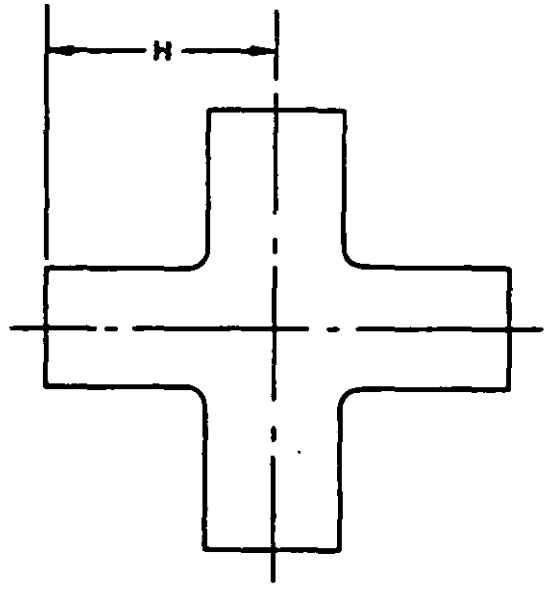
FIGURE 4. Sleeve coupling, type II.

TABLE IX. Dimensions of sleeve coupling.[1]

Nps	Overall length "G" (inches)
1-1/2	4.000
2	4.250
3	4.450
4	4.970
6	5.050
8	6.480
10	10.850
12	11.000
	11.500

[1] For tolerances, see 3.4.2.4.

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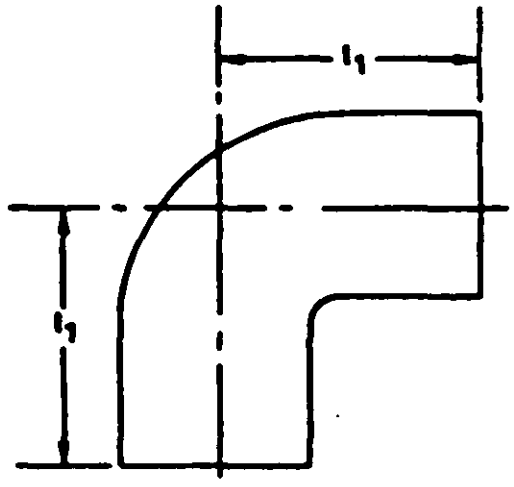
FIGURE 5. Cross, type I or II.

TABLE X. Dimensions of cross.[1]

Nps	Center to end "H" (inches)
1	2.750
1-1/2	3.375
2	3.375
3	4.625
4	5.125
6	6.125
8	11.063
10	13.125
12	14.000

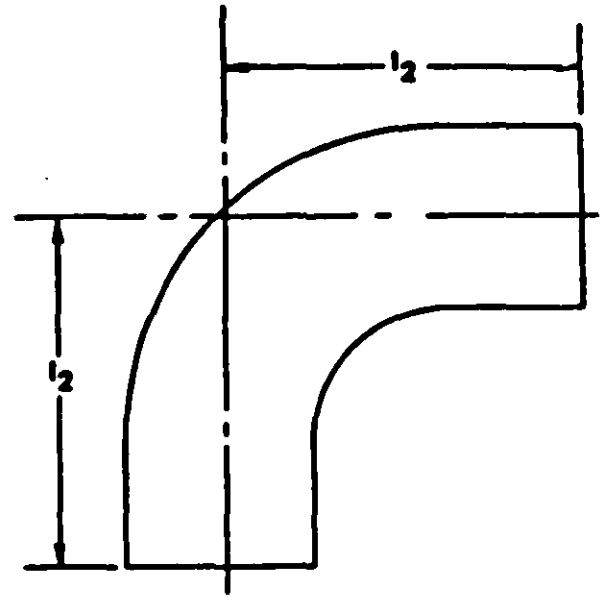
[1] For tolerances, see 3.4.2.4.

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SH 12043

FIGURE 6. 90-degree elbow, short radius, type I or II.



SH 12044

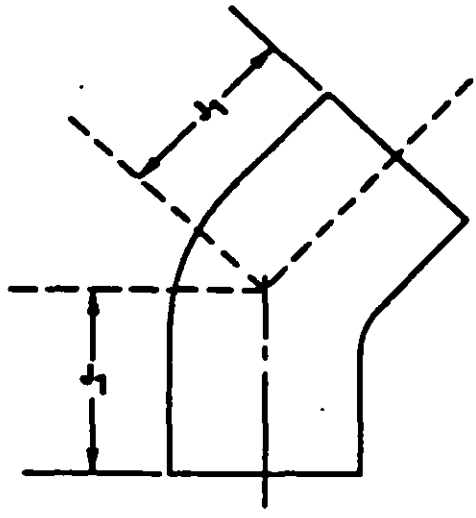
FIGURE 7. 90-degree elbow, long radius, type II.

TABLE XI. Dimensions of 90-degree elbows.[1]

Nps	Minimum center to end I_1 (inches)	Minimum center to end I_2 (inches)
1	2.750	3.750
1-1/2	3.313	4.625
2	3.375	4.565
3	4.420	6.433
4	5.125	7.263
6	6.125	8.779
8	11.625	13.000
10	13.125	15.000
12	14.000	17.800

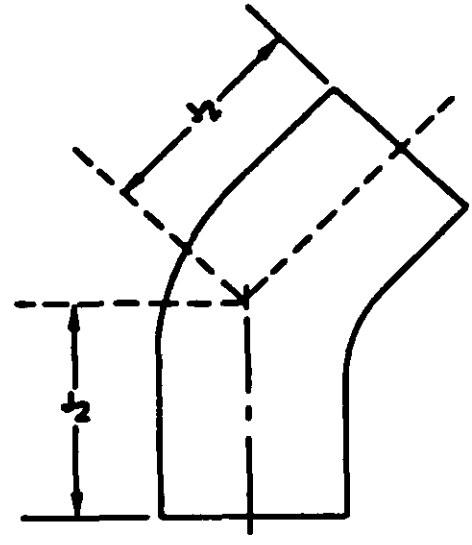
[1] For tolerances, see 3.4.2.4.

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SH 12045

FIGURE 8. 45-degree elbow, short radius, type I or II.



SH 12046

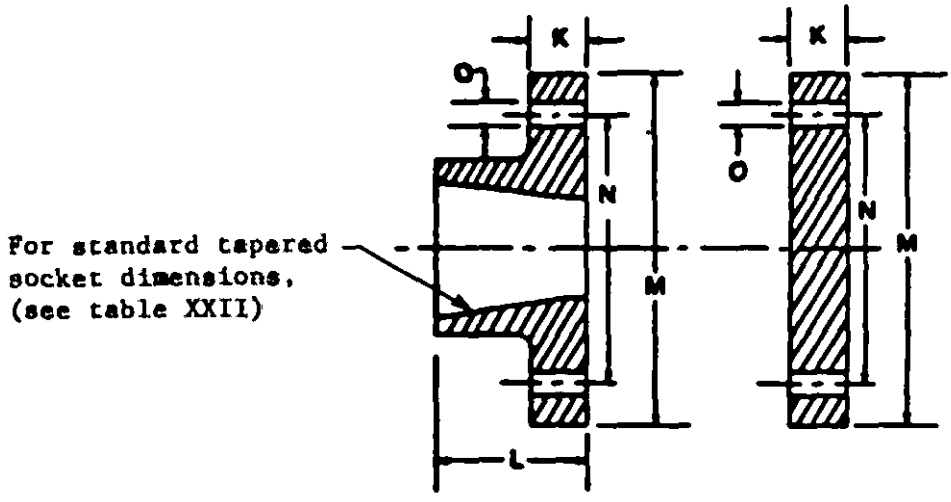
FIGURE 9. 45-degree elbow, long radius, type II.

TABLE XII. Dimensions of 45-degree elbows.[1]

Nps	Minimum center to end J_1 (inches)	Minimum center to end J_2 (inches)
1	2.375	2.188
1-1/2	2.500	2.563
2	2.625	2.625
3	3.425	3.425
4	3.875	3.875
6	4.375	4.375
8	8.125	8.125
10	8.625	8.625
12	9.500	9.500

[1] For tolerances, see 3.4.2.4.

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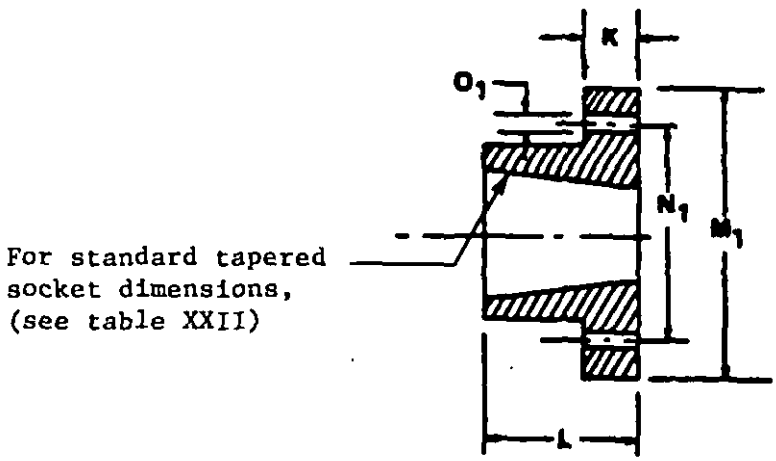
FIGURE 10. 200 lb/in² plain and blind flanges drilled to 150 lb/in² ANSI B16.5 standards, type I or II.[1]

TABLE XIII. Dimensions of 200 lb/in² plain and blind flanges drilled to 150 lb/in² ANSI B16.5 standards, including spot facing.[2]

Nps	Minimum flange thickness K (inches)	Minimum Flange width L (inches)	Outside diameter M (inches)	Bolt circle diameter N (inches)	Bolt hole diameter O (inch)	Number of bolt holes
1	0.750	1.885	4.250	3.125	0.625	4
1-1/2	.750	2.000	5.000	3.875	.625	4
2	.750	2.125	6.000	4.750	.750	4
3	1.375	2.625	7.500	6.000	.750	4
4	1.375	2.625	9.000	7.500	.750	8
6	1.500	3.000	11.000	9.500	.875	8
8	1.750	5.125	13.500	11.750	.875	8
10	2.000	5.305	16.000	14.250	1.000	12
12	2.250	5.650	19.000	17.000	1.000	12

[1] Bolting ring of plain flange may be of ribbed construction.
[2] For tolerances, see 3.4.2.4.

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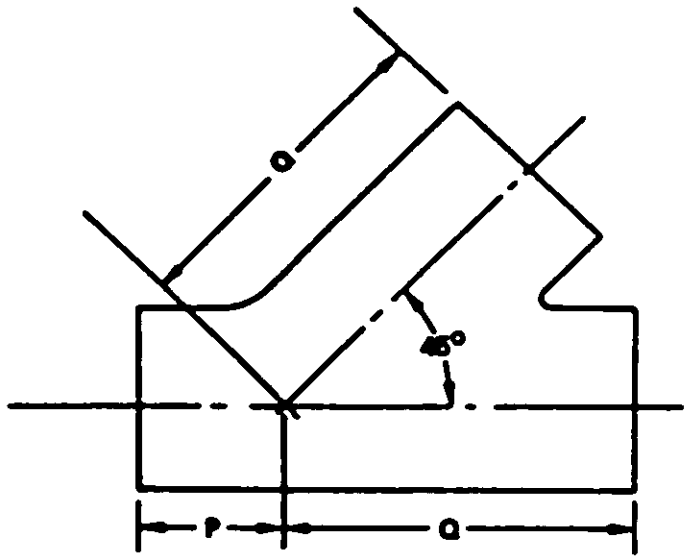
FIGURE 11. 200 lb/in² plain flange drilled to MIL-F-20042, class 150/250 pounds type II.[1]

TABLE XIV. Dimensions of 200 lb/in² plain flange drilled to MIL-F-20042, class 150/250 pounds, including spot facing. Diameter of spot facing shall be in accordance with ANSI B16.5.[2]

Nps	Minimum flange thickness K (inches)	Minimum flange width L (inches)	Outside diameter M_1 (inches)	Bolt circle diameter N_1 (inches)	Bolt hole diameter O_1 (inch)	Number of bolt holes
1	0.750	1.885	4.250	3.125	0.563	4
1-1/2	.750	2.000	5.063	3.938	.563	6
2	.750	2.125	5.563	4.433	.688	6
3	1.375	2.625	6.625	5.500	.688	8
4	1.375	2.625	7.688	6.563	.688	8
6	1.500	3.000	10.125	8.875	.688	12
8	1.750	5.125	12.375	11.063	.688	14
10	2.000	5.305	15.000	13.438	.813	15
12	2.250	5.650	17.625	16.063	.813	18

[1] Flange bolting ring shall not be ribbed.
[2] For tolerances, see 3.4.2.4.

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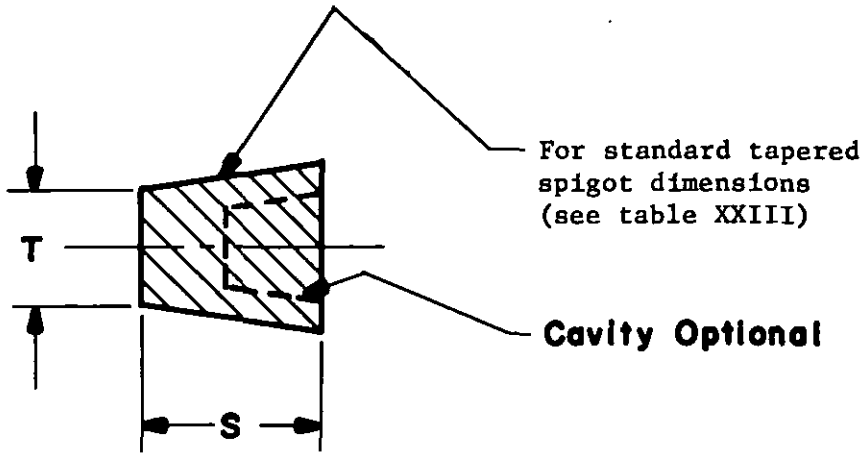
FIGURE 12. 45-degree lateral type I or II.[1]

TABLE XV. Dimensions of 45-degree lateral.

Nps	Minimum center to end (inlet) P (inches)	Minimum center to end (outlet) Q (inches)
1	2.500	3.875
1-1/2	3.250	5.250
2	2.750	6.625
3	4.250	7.750
4	4.375	9.000
6	5.750	12.500
8	7.375	16.250
10	8.750	19.625
12	11.000	24.750

[1] The lateral is rated for 50 lb/in² only.

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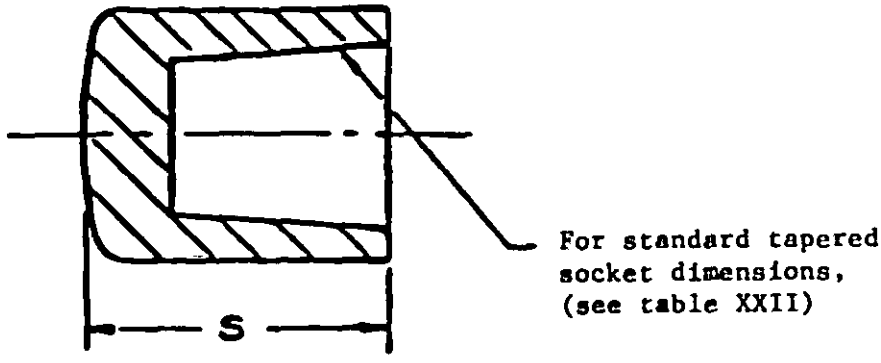
FIGURE 13. Tapered plug for fitting socket, type I or II.[1] [2]

TABLE XVI. Dimensions of tapered plug.[3]

Nps	Minimum plug length S (inches)	Minor plug diameter T (inches)	Minor plug diameter tolerance (inch)
1	1.250	1.250	± 0.010
1-1/2	1.500	1.830	± .010
2	1.500	2.290	± .010
3	1.750	3.400	± .010
4	1.750	4.405	± .010
6	2.500	6.480	± .010
8	2.750	8.490	± .025
10	3.450	10.550	± .025
12	4.150	12.570	± .025

[1] May be used with type I or II fittings.
[2] Tapered plugs may be adhesive bonded in sleeve couplings (see figure 4) to form pipe caps.
[3] For unspecified tolerances, see 3.4.2.4.

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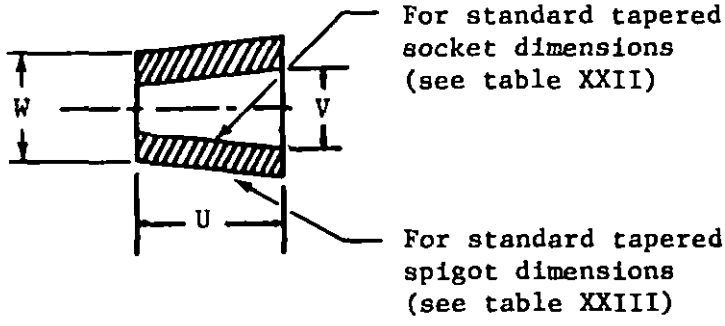
FIGURE 14. Tapered end cap for fitting male socket, type I or II.[1]

TABLE XVII. Dimensions of tapered end cap.

Nps	Cap length S minimum (inches)
1	2.250
1-1/2	2.625
2	2.750
3	3.250
4	3.750
6	4.625
8	6.500
10	7.000
12	8.750

[1] For tolerances, see 3.4.2.4.

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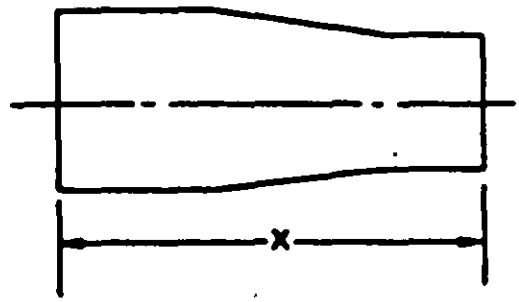
FIGURE 15. Reducer bushing, type I or II.[1]

TABLE XVIII. Dimensions of concentric reducer bushing.[2]

Nps	Minimum bushing length U (inches)	Internal minor diameter V (inches)	Internal minor diameter tolerance (inch)	External minor diameter W (inches)	External minor diameter tolerance (inch)
1-1/2 x 1	2.000	1.220	±0.010	1.830	±0.010
2 x 1	2.000	1.220	± .010	2.290	± .010
2 x 1-1/2	2.000	1.795	± .010	2.290	± .010
3 x 2	2.000	2.260	± .010	3.400	± .010
4 x 3	2.000	3.370	± .010	4.405	± .010
6 x 4	2.250	4.375	± .010	6.480	± .010
8 x 6	2.750	6.450	± .010	8.490	± .025
10 x 8	5.125	8.400	± .010	10.550	± .025
12 x 10	5.300	10.465	± .010	12.570	± .025

[1] May be used with type I or II fittings.
[2] For unspecified tolerances, see 3.4.2.4.

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SH 12052

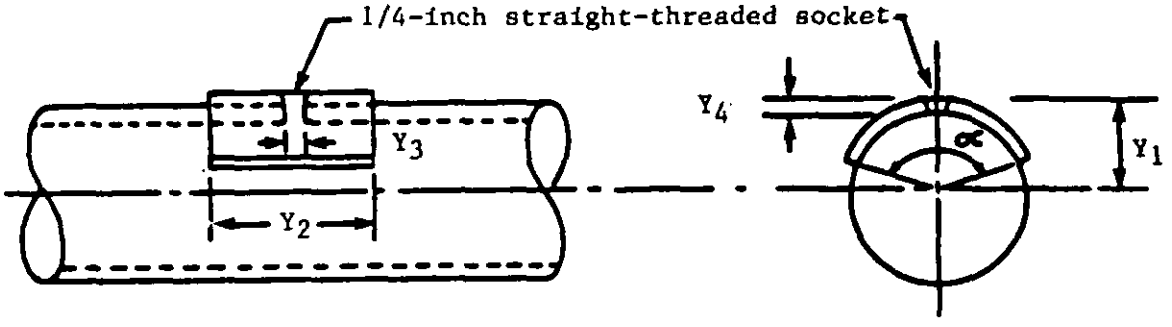
FIGURE 16. Tapered body concentric reducer.

TABLE XIX. Dimensions of tapered body concentric reducer.

Nps	Minimum overall length X (inches)
1-1/2 x 1	4.125
2 x 1	5.500
2 x 1-1/2	5.000
3 x 1-1/2	6.375
3 x 2	6.000
4 x 2	7.750
4 x 3	7.000
6 x 3	10.000
6 x 4	9.000
8 x 4	11.750
8 x 6	12.250
10 x 6	14.000
10 x 8	14.750
12 x 8	16.250
12 x 10	17.000

[1] For tolerances, see 3.4.2.4.

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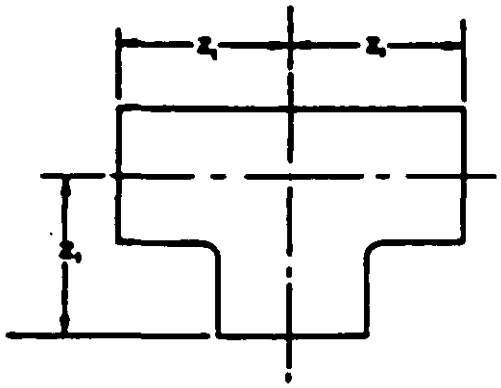
FIGURE 17. Instrumentation saddle, type I.[1]

TABLE XX. Dimensions of instrumentation saddle.[2]

Nps	Minimum height above centerline Y_1 (inches)	Saddle width Y_2 (inches)	Maximum diameter of flow passage drilled through pipe Y_3 (inch)	∝ Minimum contact angle (degrees)	Saddle thickness Y_4 (inch)
2 x 1/4	1.900	4	0.125	180	0.750
3 x 1/4	2.500	4	.125	120	.750
4 x 1/4	3.000	4	.125	120	.750
6 x 1/4	4.000	4	.125	120	.750
8 x 1/4	5.125	4	.125	60	.875
10 x 1/4	6.250	4	.125	60	.875
12 x 1/4	7.250	4	.125	60	.875

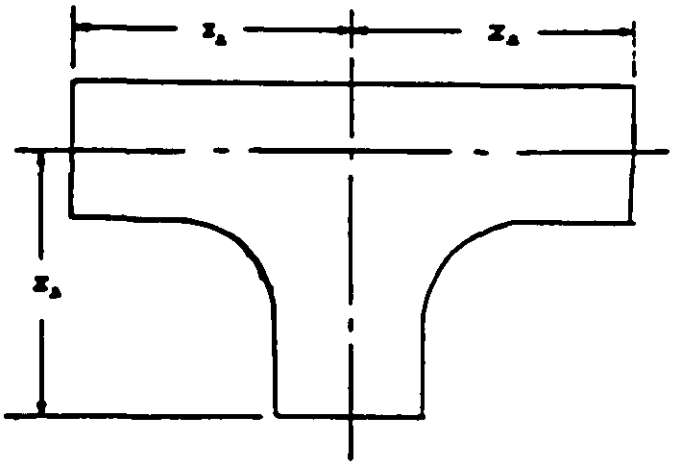
[1] After drilling flow passage through pipe wall, a thin layer of epoxy adhesive shall be applied to the freshly exposed pipe surface within the hole to protect the fiberglass reinforcement.
[2] For unspecified tolerances, see 3.4.2.4.

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SH 12054

FIGURE 18. Tee, straight short radius, type I.



SH 132317397

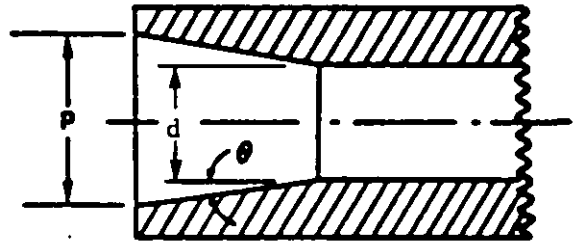
FIGURE 19. Tee, straight long radius, type II.

TABLE XXI. Dimensions of straight tee.[1]

Nps	Minimum center to end Z_1 (inches)	Minimum center to end Z_2 (inches)
1	2.750	2.375
1-1/2	3.375	2.625
2	3.375	3.938
3	4.500	4.875
4	5.125	6.375
6	6.125	7.875
8	11.625	9.500
10	13.125	11.250
12	14.000	13.000

[1] For tolerances, see 3.4.2.4.

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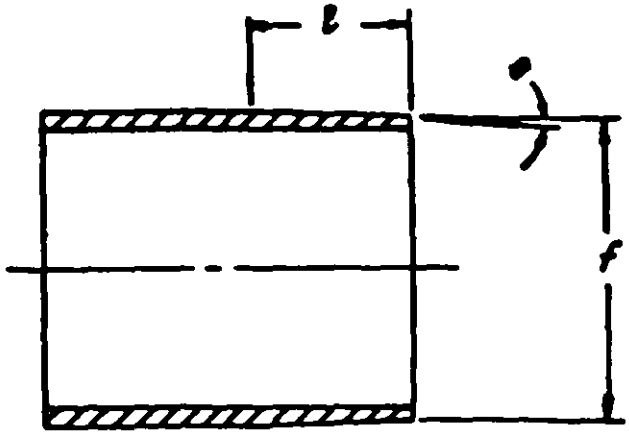
FIGURE 20. Standard tapered socket for type I and II fittings.

TABLE XXII. Dimensions of standard tapered socket.[1]

Nps	Socket entrance minimum diameter p (inches)	Socket entrance maximum diameter P (inch)	Socket minor diameter d maximum (inches)	θ Taper angle (degrees)	Taper angle tolerance (degree)
1	1.345	1.375	1.230	1.750	+.250 -0
1-1/2	1.925	1.960	1.805	1.750	+.250 -0
2	2.375	2.420	2.270	1.750	+.250 -0
3	3.500	3.540	3.380	1.750	+.250 -0
4	4.500	4.560	4.385	1.750	+.250 -0
6	6.625	6.700	6.460	1.750	+.250 -0
8	8.720	8.760	8.407	1.750	+.250 -0
10	10.800	10.860	10.746	1.750	+.250 -0
12	12.850	12.890	12.506	1.750	+.250 -0

[1] For unspecified tolerances, see 3.4.2.4.

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FIGURE 21. Standard tapered spigot for type I or II fittings.

TABLE XXIII. Dimensions of standard tapered spigot.[1]

Nps	Spigot minor diameter f (inches)	Spigot minor diameter tolerance	θ Taper angle (degrees)	Taper angle tolerance (degrees)	Minimum spigot taper length l (inches)
1	1.250	+0.010	1.750	+0.000 - .250	----
1-1/2	1.830	+ .010	1.750	+ .000 - .250	----
2	2.290	+ .010	1.750	+ .000 - .250	----
3	3.400	+ .010	1.750	+ .000 - .250	----
4	4.405	+ .010	1.750	+ .000 - .250	----
6	6.480	+ .010	1.750	+ .000 - .250	----
8	8.431 to 8.553[2]	----	1.750	+ .000 - .250	2.750
10	10.514 to 10.588[2]	----	1.750	+ .000 - .250	3.450
12	12.546 to 12.596[2]	----	1.750	+ .000 - .250	4.150

[1] For unspecified tolerances, see 3.4.2.4.
[2] This dimension varies with the od and minimum spigot taper length shall be achieved within these limits.

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